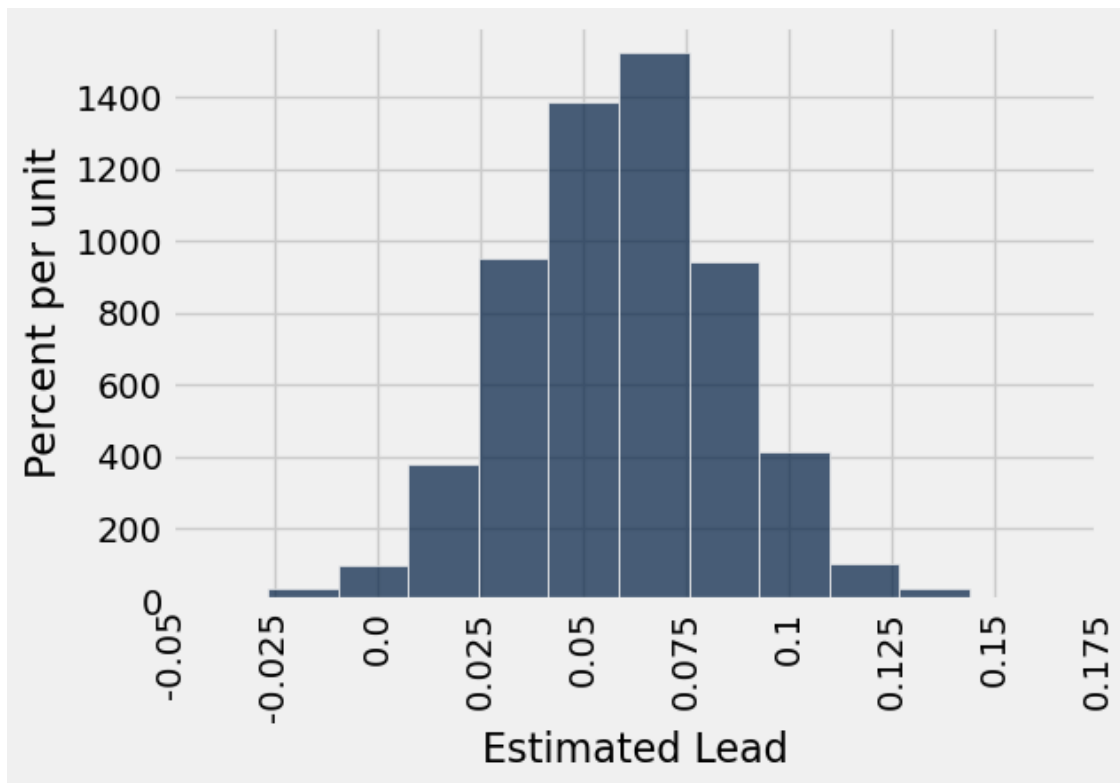


Question 1.5. Write a function called `leads_in_resamples` that returns an array of 2023 elements representing the bootstrapped estimates (the result of calling `one_resampled_difference`) of Imm Thai's lead over Lucky House, Thai Temple, and Thai Basil combined. Afterwards, run the cell to plot a histogram of the resulting samples. **(8 Points)**

Hint: If you see an error involving `NoneType`, consider what components a function needs to have!

```
In [13]: def leads_in_resamples():
    leads = make_array()
    for i in np.arange(2023):
        bootstrap = votes.sample()
        imm_percentage = np.count_nonzero(bootstrap.where('Vote', 'Imm Thai').column('Vote'))/b
        lh_percentage = np.count_nonzero(bootstrap.where('Vote', 'Lucky House').column('Vote'))
        tt_percentage = np.count_nonzero(bootstrap.where('Vote', 'Thai Temple').column('Vote'))
        tb_percentage = np.count_nonzero(bootstrap.where('Vote', 'Thai Basil').column('Vote'))/
        difference = imm_percentage - (lh_percentage + tt_percentage + tb_percentage)
        leads = np.append(leads, difference)
    return leads

sampled_leads = leads_in_resamples()
Table().with_column('Estimated Lead', sampled_leads).hist("Estimated Lead")
```



Question 2.1. Jonathan also created 70%, 90%, and 99% confidence intervals from the same sample, but he forgot to label which confidence interval represented which percentages! **First, match each confidence level (70%, 90%, 99%) with its corresponding interval in the cell below** (e.g. ____ % CI: [52.1, 54] → replace the blank with one of the three confidence levels). **Then, explain your thought process** and how you came up with your answers. **(10 Points)**

The intervals are below:

- [50.03, 55.94]
- [52.1, 54]
- [50.97, 54.99]

Hint: If you are stuck on this question, try looking over Chapters [13.3](#) and [13.4](#) of the textbook.

70% CI: [52.1, 54]

90% CI: [50.97, 54.99]

99% CI: [50.03, 55.94]

Wider intervals generally lead to higher confidence levels. The widths of the intervals [50.03, 55.94], [52.1, 54], and [50.97, 54.99] are 5.91, 1.9, and 4.02, respectively. This means that they are the widest, narrowest, and second widest intervals, respectively. So, they are matched with the highest, lowest, and second lowest confidence levels, respectively.

